P8: Exponentials and Logarithms

1. Exponential Functions

Exponential Function: A function of the form $y = a^x$ where $a > 0$, $a \neq 1$.

Key properties of $y = a^x$:

- Always passes through $(0, 1)$ since $a^0 = 1$
- Domain: all real numbers $\mathbb{R}$; Range: $y > 0$
- Horizontal asymptote at $y = 0$ (the x-axis)
- If $a > 1$: exponential **growth** (increasing)
- If $0 < a < 1$: exponential **decay** (decreasing)

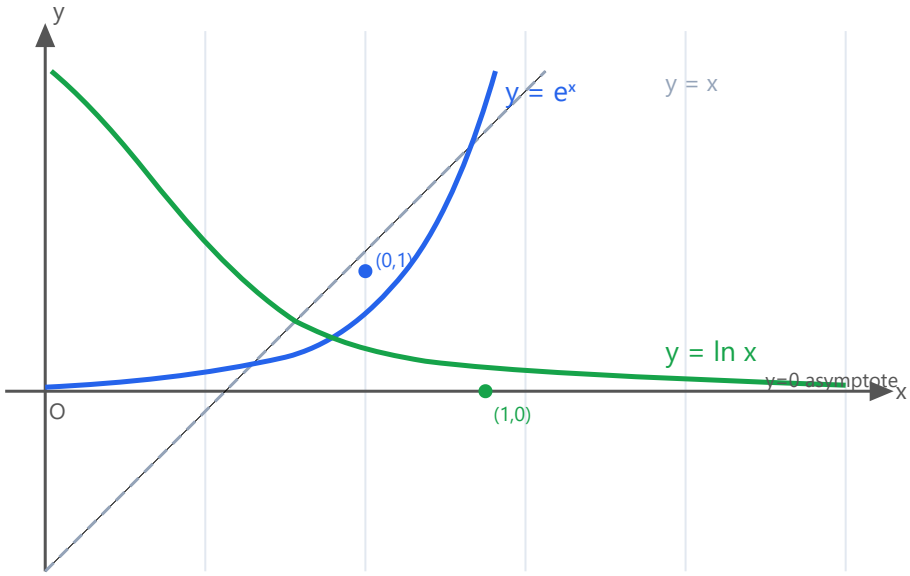
The Natural Exponential e^x

Euler's number: $e \approx 2.71828\ldots$

The function $y = e^x$ is the unique exponential function that is its own derivative:

$$\frac{d}{dx}(e^x) = e^x$$

This makes e the natural base for calculus. Also: $\int e^x \, dx = e^x + C$.



The graphs of $y = e^x$ and $y = \ln x$ are **reflections of each other in the line $y = x$** — they are inverse functions of each other.

2. Logarithms

Definition of Logarithm:

$$\log_a x = y \iff a^y = x$$

Read as: "log base a of x equals y" means "a to the power y equals x."

The logarithm answers the question: "What power do I raise a to in order to get x?"

Converting between log and exponential form:

$\log_2 8 = 3 \iff 2^3 = 8$
$\log_{10} 1000 = 3 \iff 10^3 = 1000$
$\log_5 \frac{1}{25} = -2 \iff 5^{-2} = \frac{1}{25}$
$\ln e^4 = 4 \iff e^4 = e^4$

The Natural Logarithm

The natural logarithm is $\ln x = \log_e x$. Key results:

$$e^{\ln x} = x \qquad \text{and} \qquad \ln(e^x) = x$$

These are inverses: applying one undoes the other.

3. Laws of Logarithms

Law	Formula	Example
Product rule	$\log_a(xy) = \log_a x + \log_a y$	$\log_2 6 = \log_2 2 + \log_2 3$
Quotient rule	$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$	$\log_3 \frac{9}{2} = \log_3 9 - \log_3 2$
Power rule	$\log_a(x^n) = n\log_a x$	$\log_2(8^3) = 3\log_2 8 = 9$
Log of 1	$\log_a 1 = 0$	$\ln 1 = 0$
Log of base	$\log_a a = 1$	$\log_{10} 10 = 1$
Reciprocal	$\log_a\left(\frac{1}{x}\right) = -\log_a x$	$\ln \frac{1}{5} = -\ln 5$

Common Errors to Avoid:

- $\log(x+y) \neq \log x + \log y$ — the product rule only applies to products, not sums
- $\log(x^2) = 2\log x$ only valid when $x > 0$; write $\log|x|$ if x could be negative
- $\frac{\log x}{\log y} \neq \log\left(\frac{x}{y}\right)$ — these are different things

4. Change of Base

Change of Base Formula:

$$\log_a x = \frac{\ln x}{\ln a} = \frac{\log_{10} x}{\log_{10} a}$$

This allows calculation of any logarithm using a calculator.

Calculate $\log_3 20$:
$\log_3 20 = \dfrac{\ln 20}{\ln 3} = \dfrac{2.996}{1.099} \approx 2.727$

5. Solving Exponential Equations

To solve equations of the form $a^x = b$, take logarithms of both sides.

Example — Solve $3^x = 7$:

Take $\ln$ of both sides: $\ln(3^x) = \ln 7$
Power rule: $x \ln 3 = \ln 7$
$x = \dfrac{\ln 7}{\ln 3} \approx \dfrac{1.946}{1.099} \approx 1.771$

Example — Solve $5 \cdot 2^{3x} = 40$:

$2^{3x} = 8 = 2^3$
$3x = 3 \Rightarrow x = 1$
(Or: take $\ln$ of both sides: $3x \ln 2 = \ln 8$, $x = \frac{\ln 8}{3 \ln 2} = \frac{3 \ln 2}{3 \ln 2} = 1$)

Disguised Quadratics with Exponentials

Example — Solve $e^{2x} - 5e^x + 6 = 0$:

Let $u = e^x$ (so $u > 0$ always). Note: $e^{2x} = (e^x)^2 = u^2$.
$u^2 - 5u + 6 = 0$

$(u-2)(u-3)=0$
$u=2$ or $u=3$

Case 1: $e^x=2 \Rightarrow x=\ln 2$
Case 2: $e^x=3 \Rightarrow x=\ln 3$

Exam Tip: When solving $e^{2x}-5e^x+6=0$ type problems, always substitute $u=e^x$. Since $e^x>0$ for all x, any solution $u \leq 0$ must be rejected.

6. Solving Logarithmic Equations

Example — Solve $\log_2(x+1)+\log_2(x+3)=3$:

Combine using product rule: $\log_2[(x+1)(x+3)]=3$
Convert: $(x+1)(x+3)=2^3=8$
$x^2+4x+3=8$
$x^2+4x-5=0$
$(x+5)(x-1)=0$
$x=-5$ or $x=1$

Check: $x=-5$ gives $\log_2(-4)$ — undefined. Reject.
Solution: $x=1$

Always check solutions of logarithmic equations: the argument of a logarithm must be strictly positive. Reject any solution that makes an argument zero or negative.

7. Graphs of Exponential and Logarithmic Functions

Function	Domain	Range	Asymptote	Key point
$y=e^x$	$\mathbb{R}$	$y>0$	$y=0$	$(0,1)$

$y = e^{-x}$	$\mathbb{R}$	$y > 0$	$y = 0$	$(0, 1)$
$y = \ln x$	$x > 0$	$\mathbb{R}$	$x = 0$	$(1, 0)$
$y = \ln(-x)$	$x < 0$	$\mathbb{R}$	$x = 0$	$(-1, 0)$

Transformations of e^x and $\ln x$

Standard transformation rules apply. For example:

- $y = e^{x+2}$: translation left 2 units — passes through $(-2, 1)$
- $y = e^x + 3$: translation up 3 — asymptote shifts to $y = 3$
- $y = \ln(x - 1)$: translation right 1 — asymptote shifts to $x = 1$, passes through $(2, 0)$
- $y = 2e^x$: vertical stretch factor 2 — passes through $(0, 2)$

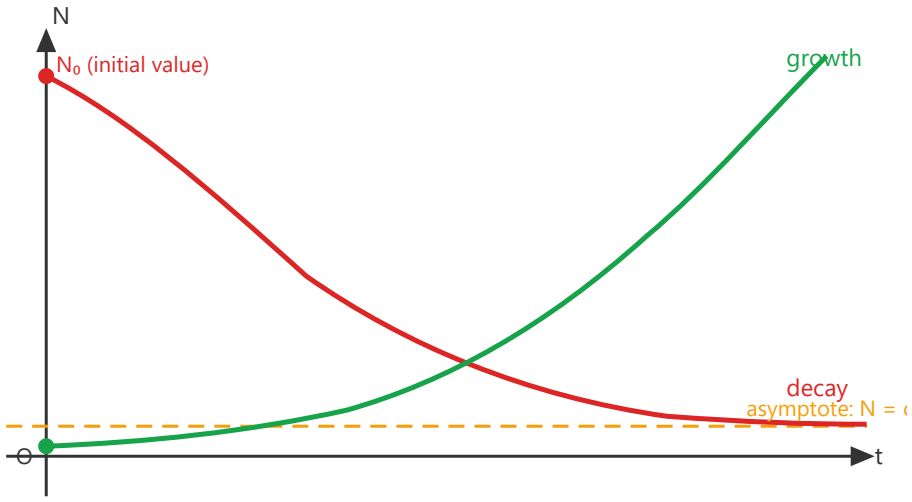
8. Exponential Growth and Decay Modelling

General Model:

$$N = N_0 e^{kt}$$

where N_0 = initial value ($t = 0$), k = rate constant, t = time.

If $k > 0$: **exponential growth**
If $k < 0$: **exponential decay**



Example — Radioactive Decay:

A substance decays according to $M = 500e^{-0.02t}$, where M grams remain after t minutes.

(a) Find the initial mass: $M = 500e^0 = 500$ g
(b) Find mass after 30 minutes: $M = 500e^{-0.6} \approx 500 \times 0.549 \approx 274.5$ g
(c) Find when $M = 100$:
$100 = 500e^{-0.02t}$
$e^{-0.02t} = 0.2$
$-0.02t = \ln(0.2)$
$t = \dfrac{-\ln(0.2)}{0.02} = \dfrac{\ln 5}{0.02} \approx 80.5$ minutes

Linearising Exponential Data

If $y = Ab^x$, taking $\ln$ of both sides gives:

$$\ln y = \ln A + x \ln b$$

Plotting $\ln y$ against x gives a straight line with gradient $\ln b$ and intercept $\ln A$. This is a common Edexcel exam technique.

Key Exam Points for P8:

- Know $e^{\ln x} = x$ and $\ln(e^x) = x$ — use these to "undo" operations
- For disguised quadratics, let $u = e^x$ and always reject $u \leq 0$
- Always check domain when solving log equations — argument must be positive
- For modelling questions, read off N_0 at $t = 0$ and use logarithms to find k or t
- Linearising: if the graph of $\ln y$ vs x is a straight line, the original relationship is exponential